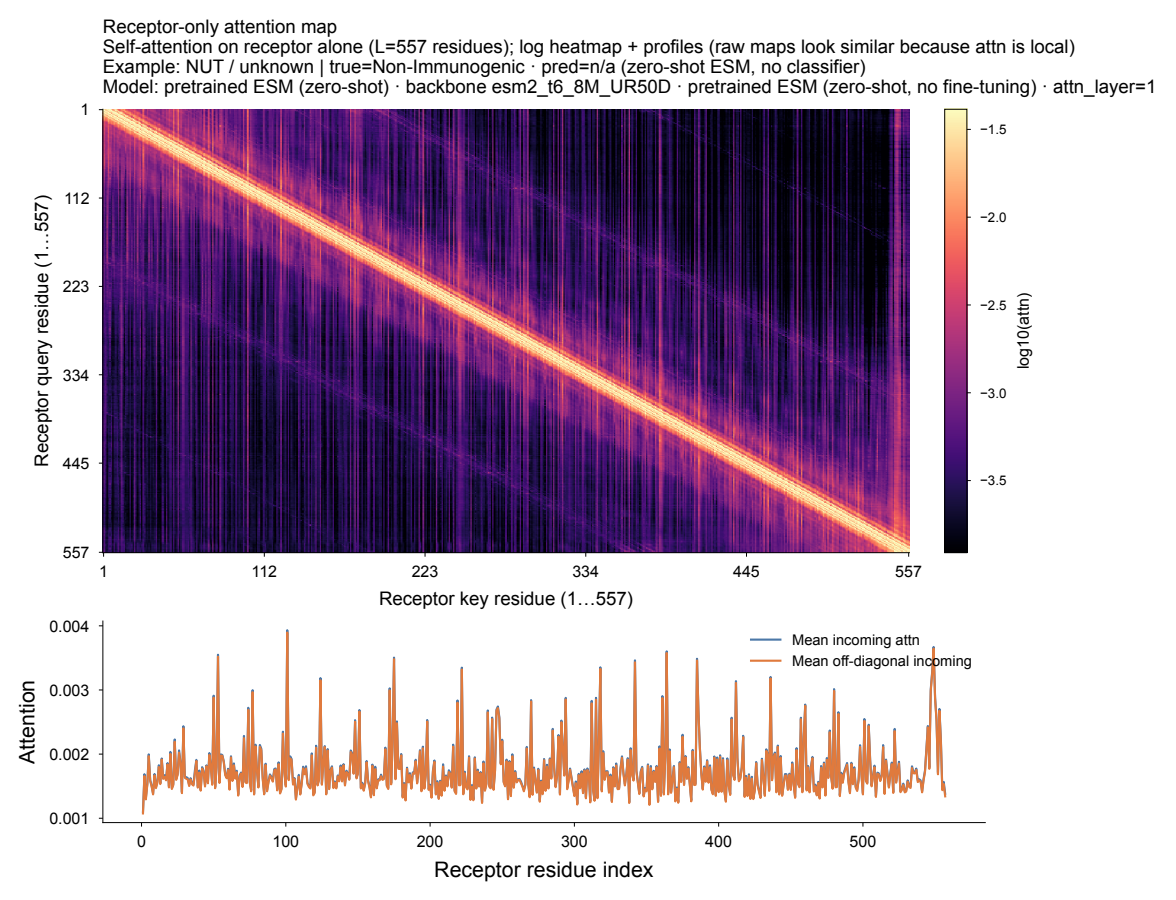

Receptor-only attention map
Self-attention on receptor alone (L=557 residues); log heatmap + profiles (raw maps look similar because attn is local)
Example: NUT / unknown | true=Non-Immunogenic · pred=n/a (zero-shot ESM, no classifier)
Model: pretrained ESM (zero-shot) · backbone esm2_t6_8M_UR50D · pretrained ESM (zero-shot, no fine-tuning) · attn_layer=1
Receptor query residue (1…557)
1
112
223
334
445
557
Receptor key residue (1…557)
log10(attn)
−1.5
−2.0
−2.5
−3.0
−3.5
Attention
0.004
0.003
0.002
0.001
Mean incoming attn
Mean off-diagonal incoming
0
100
200
300
400
500
Receptor residue index